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## MD5 Message Digest Algorithm

Aim:

To compute the 128-bit MD5 hash and verify data integrity over socket communication.

Algorithm:

Client Processing:

- \* Load openssl legacy provider.
- \* Initialize MD5 using `EVP_Digest Init -md`  
`EVP-md5()`
- \* Generate 32-character hex digest.
- \* Send message and digest.

Server Verification:

- \* Receive message & digest.
- \* Recompute MD5
- \* Compare digests.

Output:

Client:

Message : MD5 digest verification test payload

MD5 Digest : 29a8f2c3881de8d1234bc890aefd7123

Sent message & MD5 digest successfully.

Server:

Listening on port 8080

Received plaintext message: "MD5 Digest verification  
~~test~~ fast payload".

Received MD5 Digest: 29a8f2c38810c8d1234bc890aefdt123

Computed digest: 29a8f2c38810c8d1234bc890aefdt123

Integrity verified: MD5 digests match perfectly.